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On Geodesic Lines, in particular those of a Quadric Surface.

[continued from p. 166]

p. 167. (x as arbitrary parameter) as the equations of a right line on the surface; viz. considering x, p, w denoting the foregoing functions of p and q , these two equations are forms of a single relation between p, q, w , which relation expresses that the point (p, q) is situate on the right line determined by the parameter w . We may from this integral equation deduce without difficulty the foregoing differential equation; viz. we have

$$\frac{dw}{\sqrt{w}} + \frac{4dp}{\sqrt{P}} = \frac{c dp}{\sqrt{P}},$$

$$\frac{dw}{\sqrt{w}} + \frac{4dq}{\sqrt{Q}} = \frac{5dq}{\sqrt{Q}},$$

or multiplying these equations

$$\frac{dw^2}{w} + \frac{2dp dq}{\sqrt{PQ}} = 0,$$

and substituting herein for dw, dp, dq their values in terms of dp and dq , we find the required equation

$$\frac{dp^2}{(x+p)(b+p)(c+p)} + \frac{dq^2}{(a+q)(b+q)(c+q)} = 0.$$

19. I return to the integral equation involving w ; we have to rationalise this equation, that is, obtain from it an equation containing w^2, p^2, r^2 , and then substituting for these their values in terms of p, q , we have the required relation between p, q . We at once obtain

$$\left\{ 2 \left(\frac{c^2 + w}{a + b} \right) - \left(ax + \frac{r}{b} \right) (1 + cx) \right\}^2 = 4 \left(c^2 + \frac{w}{a + b} \right) (1 + cx)^2,$$

or if for greater convenience we introduce in place of a a new parameter ϕ , determined by the equation $a^2 + \frac{w}{a+b}$, the equation is

$$\left\{ 2 \left(\frac{c^2 + w}{a + b} \right) - \phi \left(ax + \frac{r}{b} \right) \right\}^2 = 4(\phi^2 - w^2 x^2) = 0.$$

Writing for shortness $p + q = X, pq = Y$, we have

$$-2bq \frac{r}{b} = a^2 + a^2 X + Y, \quad -ax + \frac{r}{b} = -bX + cY,$$

and substituting these values, the equation becomes

$$\beta(b^2 + bX + Y) = bcY = aX + Y - \phi(a^2 + a^2 X + Y) + \phi(a^2 + bcY)(b^2 + a^2 Y - aY - aXY).$$

or, what is the same thing,

$$\beta b^2 - \alpha cX - \beta a(bX - cY)^2 + \phi(-bX + cY + bXY - \alpha cX + bX^2 - cX^2 Y - aXY).$$

20. This is an equation, quadratic as regards ϕ , and also as regards ϕ ; it is of the form

$$(a + 2bp + cp^2) + 2q(b + 2bp + cp^2) + q^2(c + 2bp + cp^2) = 0.$$

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and it leads to a differential equation

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} = 0,$$

where

$$P = (b + 2bp + cp^2)^2 - (a + 2bp + cp^2)(c + 2bp + cp^2),$$

$$Q = (b + 2bq + cq^2)^2 - (a + 2bq + cq^2)(c + 2bq + cq^2);$$

and upon effecting the calculation, it is found that we have

$$P = wb^2p^2(b^2 - c^2) + 4(p + \phi)(b + p)(c + p),$$

$$Q = wb^2q^2(b^2 - c^2) + 4(q + \phi)(b + q)(c + q);$$

viz. P , Q are the same multiples of $(p + \phi)(b + p)(c + p)$, $(q + \phi)(b + q)(c + q)$ respectively; so that, omitting the common factor, or taking P , Q to represent the last-mentioned functions respectively, we have

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} = 0,$$

and since the parameter ϕ has disappeared, we see that this original equation involving ϕ is the general integral of this differential equation; viz. that the differential equation belongs to the right lines on the surface.

20. The form of the integral equation may be simplified by introducing instead of ϕ a new parameter K , connected with it by the equation

$$K = \frac{\beta b - bc + \phi c}{\beta - b + \phi}, \quad \text{or} \quad \phi = \frac{(b - K)c - bc + a}{c - K}.$$

viz. we thence deduce

$$B - a + \phi = 2aE$$

$$\beta b - cn + \phi c = 2aKE$$

$$\phi(a^2 + bw)^2 - \phi(a^2 + bw)w + \phi = (K^2 + a^2K + bcK + bw)c(c - K),$$

$$\phi + q = 4K(K - b)(K - c),$$

$$\phi + r = (b - K)(c - K),$$

where denoting $(c + x)^2$ and substituting these values, the conditions are satisfied: we thus again see δ patient that the right lines $x = \text{const.}$ and $r = \text{const.}$ are geodesics.

20. The other values of the radicals $\sqrt{P}$, $\sqrt{Q}$ must present itself with a negative sign; and we thus have dp , dq each expressed in terms of da , dr ; viz. we have the equations

$$\frac{dp}{\sqrt{P}} - \frac{dq}{\sqrt{Q}} = 0,$$

and instead of K introduce the parameter $C = b - K$, the equation becomes

$$\{-p - bcc^2 + 2(C - a)(C - bw)^2 + (a + c)(C(c^2 + Y)) = 0,$$

and from these equations $\phi = \beta = \theta + (a^2 - a^2b)(p + c)(p^2 + p)$ obtain

$$(p + q)(b + Y)(c + Y)Y^2 = pq.$$

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or expanding and reducing, this is

$$\begin{aligned} & (Y + (b - C)X)^2 \\ & + Y(a - 2k - 4bC + 4C^2) \\ & + X(2aC - 4C - 2c)C \\ & + a^2 - 4aC + 4 \end{aligned}$$

or say

$$\begin{aligned} & p^2 - 4aC \\ & + (3aA - bC + 2C^2)(p+q) + p(c(2a+4bC^2-8C^2)pq + (4-4C^2)(pq)^2 + 2pq + 2(A-C)(p+q) + pq^2) = 0, \end{aligned}$$

viz. this, examining the constant C , is the general integral of the differential equation

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} = 0,$$

where

$$\begin{aligned} P &= (a+p)(b+p)(c+p), \quad a = ap + bp^2 + cp^2, \\ Q &= (a+q)(b+q)(c+q), \quad q = aq + bq^2 + cq^2. \end{aligned}$$

21. The constant C is connected with the parameter w , which originally served to determine the right line, by the equation

$$w = \frac{1}{b} \cdot \frac{2(b-c)(b-C) - Cb}{b - Cb} \cdot \frac{Cb}{c - C}$$

or, what is the same thing,

$$w = \frac{1}{b-c} \cdot \frac{2c^2 - C - C(2-a-b)}{C - b - C}.$$

Reverting to the equation between p, q, ϕ , I remark that if ϕ be therein considered as variable, we have the differential equation

$$\sqrt{q} dp + \sqrt{q} d\theta = 0,$$

where P, Q have the foregoing values

$$P = wb^2p(b^2 - c^2) + 4(p + \phi)(b + p)(c + p), \quad Q = ko,$$

and where, if the integral equation be written in the form

$$L + 2M\phi + N\phi^2 = 0,$$

then we have $\Phi = M^2 - LN$, so that

$$\Phi = b\phi^2[p\phi(p+q) + p^2(p+q) + q^2(p+q) + p^2 + q^2].$$

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22. Changing the notation, and writing

$$\begin{aligned} P &= (a + p)(b + p)(c + p), \\ Q &= (a + q)(b + q)(c + q), \\ \Phi &= (\phi + p^2)(a + b - 2c - \phi), \end{aligned}$$

the equation is

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} + \frac{\sqrt{2} d\phi}{\sqrt{\Phi}} = 0;$$

or if instead of ϕ , we introduce the original parameter w , then, observing that

$$\frac{2dw}{b} = \frac{d\phi}{\sqrt{w(a + b - w)}}$$

we at once find

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} + \frac{4dw}{\sqrt{w}} = 0,$$

where

$$\Sigma = w(1 + w) - 2(a + b - 2c)w + 4a^2;$$

or, what is the same thing,

$$\Sigma = a(w^2 - 1)^2 - b(w^2 + 1)^2 + ac; 4ac^2;$$

viz. passing from a point (p, q) on the line w to a consecutive point $(p + dp, q + dq)$ on the line $w + dw$, the above is the relation between the variations dp, dq, dw . If ϕ be the parameter of the other line through the same point, then we have in like manner,

$$\frac{dp}{\sqrt{P}} - \frac{dq}{\sqrt{Q}} + \frac{4dw'}{\sqrt{w'}} = 0,$$

(viz. one of the radicals $\sqrt{P}, \sqrt{Q}$ must present itself with a reversed sign); and we thus have dp, dq each expressed in terms of dw, dw' ; viz. we have the increments dp, dq when a point passes from (w, w') to $(w + dw, w' + dw')$. These results will be presently obtained in a more simple manner.

Formulae where the position of a Point on the Surface is determined by means of the two Lines through the Point.

23. We may determine the position of a point by means of the parameters w, r of the two lines through the point. The equations of these are

$$\begin{aligned} \frac{x}{\sqrt{a}} &= \frac{Cw}{\sqrt{b}} = \left(1 - \frac{1}{w}\right) \frac{z}{\sqrt{c}} \quad \left(1 - \frac{Cw}{x}\right) \frac{w}{\sqrt{a}}, \\ \frac{x'}{\sqrt{a}} &= \frac{Cw'}{\sqrt{b}} = \left(1 - \frac{1}{w'}\right) \frac{z}{\sqrt{c}} \quad \left(1 - \frac{Cw'}{x}\right) \frac{w}{\sqrt{a}}, \end{aligned}$$

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and from these equations we deduce

$$\frac{x}{\sqrt{a}} = \frac{aw + Y}{a + w}, \quad \frac{Cy}{\sqrt{b}} = \frac{cw - 1}{c + w}, \quad \frac{z}{\sqrt{c}} = \frac{Y - aw}{Y + w}.$$

We have therefore

$$\begin{aligned} \frac{dx}{\sqrt{a}} &= (w^2 - 1) dw + (w^2 - 1) dr \quad (+), \\ \frac{bdy}{\sqrt{b}} &= (w^2 + 1) dw + (w^2 + 1) dr \quad (-), \\ \frac{dz}{\sqrt{c}} &= -B dw + 2aw dr \quad (+), \end{aligned}$$

where denoting $(c + a)^2$, regarding w, r as the parameters in place of p, q , then those the values of the first differential coefficients $\frac{dx}{dw}$, etc. are

$$A = -B\sqrt{a}(2w - 1), \quad B = -B\sqrt{a}(2w + 1), \quad C = -B\sqrt{c}(2 - 2w),$$

where denom. = $(r + w)^2$. We have, moreover,

$$\begin{aligned} \frac{d^2x}{dr^2} &= -2(w' - 1) dr^2 + 2(w' + 1) du - 2(w' - 1) dr^2 \quad (+), \\ \frac{bd^2y}{dr^2} &= -2(w' - 1) dr^2 + 2(w' + 1) du - 2(w' - 1) dr^2 \quad (-), \\ \frac{d^2z}{dr^2} &= 4w dw + 2(1 - w) du \quad (+), \end{aligned}$$

where dn. = $(r + w)^3$. We have, moreover,

$$\begin{aligned} \frac{d^2x}{dr^2} &= (w' - 1) du - (w' + 1) du^2 \quad (-), \\ \frac{bd^2y}{dr^2} &= (w' - 1) du - (w' + 1) du^2 \quad (-), \\ \frac{d^2z}{dr^2} &= -4w dw + (1 - w) du \quad (-), \end{aligned}$$

that is, $E' = w$ and $G' = 0$, on the differential equation of the chief lines is $dwd r = 0$, which is right. The value of E ($= dx^2 + dy^2 + dz^2$) is readily obtained, but it is rarely required; but it is easily found that

$$EG - F^2 = (EF + YB)(r + w)^2(w + r)^2 + r^2 + w^2;$$

or since the term in $\sqrt{(\quad)} = -(r + w)^2$, we have

$$\frac{F}{c + r} = \frac{-w(c + a)w}{(r + w)^2}.$$

24. The values of E, F, G ($dx^2 - 2dxdr + 2drdx + 2drdx + Gdr^2$) are

$$\begin{aligned} E &= a(c + r)^2 & + b(c - 1)^2 + c(2w - 1)^2 &= 4w, \\ F &= -a(w^2 - 1)(w'^2 + 1) & - b(w^2 - 1)(w'^2 - 1) + 2c(2w - 1) dw &= 4w, \\ G &= a(c + w')^2 & - b(c - 1)^2 + c(2w - 1)^2 &= 4w, \end{aligned}$$

where denom. = $(r + w)^2$.

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We have, it is clear, $(E_1 + d_1K = K_1 + d_1G)$, that is

$$K_1 = \frac{4}{r+w} E, \quad G_1 = \frac{4}{r+w} G.$$

Hence the condition $FG_1 - 2F_1G + GF_1 = 0$, in order that $w = \text{const.}$ may be a geodesic, reduces itself to

$$\frac{1}{r+w} F = 2F_1 + G_1 = 0,$$

and similarly, the condition $-2EFG + E_1F + EFG = 0$, in order that $r = \text{const.}$ may be a geodesic, reduces itself to

$$-\frac{1}{r+w} F - 2F_1 + E_1 = 0.$$

We have at once

$$\begin{aligned} E_1 &= 4a(w-1)(w+1) &= -4b(w'+1)(w-1) + \&c. \ 2c(w-1)dw, & (+). \\ G_1 &= 4a(w+1)(w+1) &= -4b(w'+1)(w'-1) + \&c. \ 2c(w+1)dw, & (+). \\ F_1 &= 2c(w'+1)(w'+1) &-a^2(c-1)(w'-1)dw = 2c(w+1)dw, & (+). \\ F_1 &= 2c(w'+1)(w'+1) &-a^2(c-1)(w'-1)dw = 2c(w+1)dw, & (+). \end{aligned}$$

where denom. $= (c+w)^2$; and substituting these values, the conditions are verified: we thus again see δ posteriori that the right lines $w = \text{const.}$ and $r = \text{const.}$ are geodesics.

25. The last-mentioned values of E, G are $E = Y(c+w)^2, G = 2(c+w)^2$, and setting for a moment

$$A = (1+r^2)(w+1) - 1)(b-1) + 1)(b+1) - 4cw,$$

we have $F = A - (b+w)^2$, the value of dz^2 , viz.

$$ds^2 = Ydr^2 + 2Adrdw + 2dw^2 - (b+w)^2dw^2,$$

which should be

$$= Ydr^2 + 2Adrdw + 2dw^2 - (b+w)^2dw^2,$$

where, as before,

$$P : Q = (a+p)(b+p)(c+p), \quad (a+q)(b+q)(c+q),$$

respectively. We have already found

$$\frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} + \frac{4dw}{\sqrt{w}} = 0,$$

or, what is the same thing,

$$\frac{dp}{\sqrt{P}} - \frac{dq}{\sqrt{Q}} + \frac{4dw'}{\sqrt{w'}} = 0;$$

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and we ought therefore to have identically

$$(p - q)^2 \left[p \left(\frac{dr}{\sqrt{P}} + \frac{dw}{\sqrt{Q}} \right)^2 - 1 \right] \left(\frac{dr}{\sqrt{P}} - \frac{dw}{\sqrt{Q}} \right)^2 = Ydr^2 + 2Adwd r + 2dw^2 - (r + w)^2 dw^2,$$

that is

$$(p - q)^2 = Y(1 + r)^2(b + w)^2 + 2 = 4Y(1 + r)^2(b + w)^2,$$

or, what is the same thing,

$$(p - q)^2 = -A^2 \sqrt{PQ} = (r + w)^2,$$

which are easily verified.

26. In fact, the equation

$$\frac{r^2}{a + b} + \frac{c^2}{a + b} + \frac{w^2}{a + b} = 1$$

gives for a quadric equation, the roots of which are $w = p, w = q$; that is, we have

$$\begin{aligned} p + q &= a^2 + r^2 a + r - b - r, \\ pq &= -((1 - b)(1 - w))^2 - (a + b + c)(1 + r^2) + ba + 1 - rb; \end{aligned}$$

and substituting herein for a^2, β^2, γ^2 , their values in terms of a, β, γ , we find

$$\begin{aligned} p + q &= -(a - c)(r^2 + 1) - (a + c)(r + r^2) - 4cw &= (r + w)^2, \\ p + q &= (ac)(r + w) - (a + c)r^2 - (a + c)(r + 2w) - 4cwr &= (r + w)^2, \end{aligned}$$

the first of which is, in fact, $p + q = -A + (r + w)^2$. And from the two equations, forming the combination $(p + q)^2 - 4pq$, we at once obtain the other equation

$$(p - q)^2 = Y(r + w)^2.$$

27. The most ready way of obtaining the relations between the differentials of p, q, w, r , is from the foregoing expressions of $p + q, pq$. Writing for a moment $p + q = -A + (r + w)^2$, $pq = D + (r + w)^2$, we have $p^2 + q^2 = -2A + (r + w)^2$, and from the first of these equations we

$$p^2(x + r^2) + p^2(x - 1)(b + 1) + p^2(x + 1)(b - 1) = 4 + (r + w)^2,$$

which is quadratic in p, w, r . The negative dimensions are

$$-1 \cdot 1 + a + b + bw + (r + w)^2,$$

$$-2 \cdot a + b - (r + 1)(r + 1)(c - 1) - 2cwa - r(c + 1) + cb;$$

respectively in three other terms; that is, three are these positions of the point P , and of course three corresponding positions of the point P' .

28. In the form just given the equation introducing w, r instead of p, q , the equation becomes

$$\sqrt{(p + r)(b + p)(c + p)} \left(\frac{dw}{\sqrt{w}} + \frac{dw}{\sqrt{r}} \right) + \sqrt{(q + r)(b + q)(c + q)} \left(\frac{dw}{\sqrt{w}} - \frac{dw}{\sqrt{r}} \right) = 0;$$

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or, what is the same thing,

$$p(q+b) \left(\frac{dw}{\sqrt{w}} + \frac{dw'}{\sqrt{w'}} \right)^2 - q(p+b) \left(\frac{dw}{\sqrt{w}} - \frac{dw'}{\sqrt{w'}} \right)^2 = 0;$$

that is,

$$(p-q) \left\{ \frac{dw^2}{w} + \frac{dw'^2}{w'} \right\} + 2(b(p+q) - 2pq) \frac{dw dw'}{\sqrt{w} \sqrt{w'}} = 0;$$

or substituting herein for $p-q$, pq , $p+q$ the values $\sqrt{Y} R$, A , such divided by $(r+w)^2$, this is

$$\sqrt{Y}(R dw^2 + R dw'^2) + 2\{b(A-2)A\} \frac{dw dw'}{\sqrt{w} \sqrt{w'}} = 0,$$

viz. writing herein $S = 0$, the equation is $dw dr = 0$, giving the right lines on the surface; and any solution of $S = 0$ gives a geodesic on the surface.

29. The equation $dw' = S dw' + 2A dw dw' + 2dw^2 - (r+w)^2 dw' = 0$ gives the right lines $w = \text{const.}$, $w' = \text{const.}$ on the case of an umbilicus form on the parabolicaloids, the value whereof are $\sqrt{Y} dw = (a+1)dw'$ and $\sqrt{Y} dw' = (b+1)dw$, that is, the ratio of the coefficients of dr , dr' is of the form function $w = \text{const.}$; and it thus appears that it is possible to draw on the surface the two sets of right lines, the lines of such not being such umbilicus that the number is divided into parallelograms, the sides of which have to each other any given ratio (the angles being variable); viz. if this ratio be at all = 1, then, to determine the relation between w , r , we must have $\sqrt{Y} dw = \pm ndu$; or what is the same thing, $\frac{dw}{\sqrt{w}} = \pm \frac{dr}{\sqrt{r}}$. In particular, if $m = 1$, the parallelograms will be rhombs; and we must then have

$$\frac{dw}{\sqrt{w}} = \pm \frac{dr}{\sqrt{r}},$$

viz. this being in terms of w , r , the differential equation of the curves of curvature, it appears that the two sets of lines may be taken so as to divide the surface into infinitesimally small rhombs; such fact, showing the diagonals of these are the two sets of curves of curvature.

The Ellipsoid and the three Hyperboloids.

30. I have thus far considered a quadric surface in general; the various theorems being applicable as well to the ellipsoid and the hyperboloids of two sheets as to the sheet hyperboloid, the right lines being of course imaginary for the first-mentioned surfaces, but I will now consider the ellipsoid and the three hyperboloid separately.

31. First the ellipsoid. We have here a , b , c all positive, and I assume as usual $a > b > c$. The principal sections are all ellipses, viz. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is the major-

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mean, or say the minor section, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ the minor-mean, or say the major section, and $\frac{x^2}{a^2} + \frac{y^2}{c^2} = 1$ the minor, or umbilicar section. The elliptic coordinates p, q enter into the equations symmetrically; but we distinguish them by taking b extend from $-c$ to $-b$, and q to extend from $-b$ to $-a$. Thus $p = \text{const.}$ denotes the curves of

[Figure: ellipse with axes labelled — omitted]

curvature of the one kind; viz. $p = -c$ denotes the major-mean section $\frac{x^2}{a^2} + \frac{y^2}{c^2} = 1$, $p = -b$ the section UU' and $V'V'$ of the umbilicar section; and $q = \text{const.}$ denotes the curves of curvature of the other kind, viz. $q = -b$ the remaining sections UVU' and $U'V'$ of the umbilicar section, $q = -a$ the minor-mean section, and $q = \text{const.}$ the minor-mean section.

32. Hence, in order that the geodesic line $\frac{dp}{\sqrt{P}} \pm \frac{dq}{\sqrt{Q}} = 0$ may have a meaning,

$$dr \int \frac{dr}{\sqrt{(a+r)(b+r)(c+r)(p+r)(\phi+r)}} \pm dq \int \frac{dq}{\sqrt{(a+q)(b+q)(c+q)(p+q)(\phi+q)}} = 0$$

of the geodesic lines may be real (observing that we have $a+p, b+p = +, c+p = +, p+q = +$, and $a+q = +, b+q = +, c+q = +$ necessarily; $p+\phi = +, q+\phi = -$ as the case may be; $p+q = +$ if included between the limits a, c . Or, what is the same thing, ϕ is included between $-b$ and $-c$, this point line is a triloid.

[Figure: pair of conics on the ellipsoid — omitted]

in the annexed diagram the values of $-p, -q, -P$. Hence on the ellipsoid we have two kinds of geodesic lines, and three leading a vast variety of consecutive curves; the geodesic line $\phi = \text{const.}$ may be means in two ways, viz. for points above the umbilicar curve, p is final between $-b$ and $-c$, but for points below the umbilicar curve, q is between $-b$ and $-a$, where this latter case the constant value of ϕ is included within a quadrant of the Royal Soc., vol. xxx., pp. 25–32, 1872, [478].

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33. Next, for the above hyperboloid, we have a and $b = +$, $c = -$, and I assume for convenience $a > b$. Attending to the signs, we still have therefore $a > b > c$. The principal sections are: one of them an ellipse, and the other two hyperbolas, viz. the minor section is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the major section is the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{c^2} = 1$, and the mean section is the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{c^2} = 1$. There are no umbilici. The elliptic coordinates enter symmetrically; but as before, we distinguish them, viz. we take p to extend from $-c$ (a positive value greater than a); but for sections of the hyperboloid, the parameters of which are between $-c$ and b , see; on the parts of the surface surrounding the hyperboloid, and q a constant.

The curves of curvature of the one kind, $p = \text{const.}$, are real; the curves of curvature of a hyperbola two; we may say that $p = \text{const.}$ are the elliptic curves and curves of curvature, that is, $-b$ and $-c$. Thus on $-b$ and $-c$ the curves of curvature of a hyperboloid the cone, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and not the intermediate values gives a curve of curvature of a hyperboloid: thus we may say that $p = \text{const.}$ determines the real curves of curvature, and the hyperbolic curves of curvature.

[Figure: section — omitted]

and in the fourth case it has a q -value; but in the second and third cases it has neither a p -nor a q -value. This is better seen from the diagram. It follows that we have, on the hyperboloid, geodesic lines of four different kinds: these which touch a curve of curvature, on a hyperboloid, and these which touch one of the two kinds of curvature, but in which a foot a positive value from 0 to $-c$, but neither curves which are negative values from $-c$ to $-a$.

35. To explain this more in detail, consider the geodesic which start from a point M of the hyperboloid. In the first instance, consider the axis of x rotation, the discussion will be much like that of the geodesic of an ellipse.

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The geodesic of initial direction $M1$ touches at M the oval curve of curvature M_2 , and lies wholly above this curve; it makes an infinity of convolutions round the upper part of the hyperboloid, cutting all the oval curves of curvature for which p has a (positive) value greater than p_1 , (if p is the value of p corresponding to the oval curve through M), and according to whether it is considering the curves as described in the opposite sense, it descends from infinity to touch the oval curve through M , after which it again ascends to infinity.

[Figure: hyperboloid sections with geodesics through M — omitted]

Next, if the initial direction is $M2$, we have a geodesic of the same kind, only descending below M to touch a certain oval curve having for its parameter p_2 , ($p_2 > -c$, $> p_1$).

We come next to a critical direction $M3$, for which the geodesic descends below M to touch the oval curve of curvature $p_2 = -c$, that is, the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. But it is to be observed that, whatever the initial point M may be, the geodesic makes below M an infinity of convolutions round the hyperboloid, so that it does not in fact ever actually touch the ellipse, but has this ellipse for an asymptote. That this is so appears from the consideration that the ellipse, just above the curvature of curvature, is a geodesic; so that, starting from a point of the ellipse in the direction of the ellipse, the geodesic coincides with the ellipse, or, besides the ellipse itself, there is not on the geodesic which touches the ellipse.

Next, if the initial direction be $M4$, the geodesic does not have touch any oval curve; it descends through M below the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, lying in the upper and lower portions of the hyperboloid, and making round it an infinity of convolutions.

36. We come, then, to the initial direction $M5$, which is that of the right line, the geodesic here coincides with the right line.

In the case which follow, the geodesic lies in the upper and lower portions of the hyperboloid, cutting all the oval curves of curvature.

Initial direction $M6$: the geodesic does not touch any hyperbolic curve of curvature, and it must touch the hyperboloid as making of convolutions.

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Initial direction *M7*: the geodesic touches at opposite infinities the mean hyperbolic $\frac{x^2}{a^2} - \frac{y^2}{c^2} = 1$, lies wholly in front of the plane $y = 0$ of this hyperbola.

Initial direction *MS*: the geodesic touches a hyperbolic curve of curvature, parameter q where q , (negative) is between $-b$ and $-q$, the parameter of the hyperboloidal curves of curvature lying between $-b$ and $-q$, but does not cut the remaining curves the parameter of which extend from q to $-a$.

Lastly, initial direction is *M9*, that of the hyperbolic curve of curvature through M ; the geodesic touches this curve, cutting all the hyperbolic curves the parameters of which are between $-b$ and q_1 , but not any of those the parameters of which are between q_2 and $-q$.

37. If in the differential equation of the geodesic line we consider p, q as the elliptic coordinates of a given point M of the curve, the equation between the variables of θ determines the direction of the curve; or conversely, if the direction be given, the equation determines the value of the parameter θ . Writing

$$\frac{dp\sqrt{p}}{\sqrt{(a+p)(b+p)(c+p)}} = P, \quad \frac{dq\sqrt{q}}{\sqrt{(a+q)(b+q)(c+q)}} = Q,$$

then P, Q are proportional to the rectangular coordinates of a consecutive point M' measured from M in the directions of the hyperbolic and oval curves of curvature respectively; and the differential equation of the geodesic line gives

$$\frac{P}{\sqrt{p+\theta}} + \frac{Q}{\sqrt{q+\theta}} = 0;$$

viz. if ϕ be the inclination of the geodesic to the hyperbolic curve of curvature, then $Q = P \tan \phi$, or we have $\frac{\tan \phi}{p+\theta} + \frac{1}{q+\theta} = 0$; that is, $p \tan^2 \phi + q + \theta(1 + \tan^2 \phi) = 0$; hence, for the right side $\phi = 0$, then $q + \theta = 0$, and therefore $\theta = -\frac{(p \tan^2 \phi + q) \tan^2 \phi}{1 + \tan^2 \phi}$; viz. $\theta = q + \theta = -p, \theta = -q$ where, as it should be.

[509]

Plan of a Curve-Tracing Apparatus.

*[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 345–347.
Read May 9, 1872.]*

p. 179. I HAVE devised a curve-tracing apparatus on the following plan:

Imagine two planes II, II' moving in the same horizontal plane, and above the two planes respectively the two points P, P' moving in the same or a specified plane.

To fix the ideas, suppose that the two planes each move according to a law (that is, let the motion of each of these depend on a single variable parameter; for instance, they may each of them rotate about an axis); but let the motion of the two points be free.

Suppose, next, that the planes are connected together in such manner that the motion of one of them determines the motion of the other (e.g. by a train of wheel-work); and that the two points are also connected together in such manner that the motion of one of them determines the motion of the other (e.g. by a pantograph; or by a slotted rod, the slot of which works on a pin, in like manner as the rod or move lengthways as well as rotate).

Suppose, finally, that one of the points, say P , is attached to a point of the plane II', then the plane II being set in motion, this determines the motion of II', of P in a relative curve $ABCDEFGFG$, with through these ten points. (Or ABC , is derived from the points A, B, C ; and in like manner the points $a, b, c, d, e, f, g, h, i, j$ of in mere lengthways as well as rotate.)

I propose to describe the apparatus as nearly as I have actually constructed it. (See sketch-plan Fig. 1, and side-elevation Fig. 2.)

The framework of the instrument consists of two longitudinal bars (B) each about three feet long, one inch thick, and three inches broad, supported together at a distance of about eighteen inches on the cross-pieces C, C' and half-way between them, supported by the same cross-pieces, is an axis carrying at each extremity two mitre wheels. The bars B support three cross-pieces D, D, D , and between these are

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the moveable cross-pieces E, E carrying the axes of A, A with the attached mitre wheels, and circular disks X, X . Each of these wheels separately may be placed (as in the figure) out of gear with the two central wheels, or it can (by moving the moveable cross-pieces) be put into gear, and so be turned by them.

The disks X, X may be regarded as being themselves the planes II, II' (any planes are rigidly attached to X, X respectively); or we may, in any other manner, move the plane II by means of the disk X , for instance X may carry a spur-wheel gearing into a spur-wheel on the under surface of II , and thereby communicating to it

[Figure: Diagram of curve-tracing apparatus, plan and elevation — omitted]

a rotation of different velocity to the plane II ; or the connexion may be as in the ordinary rod chuck. In any such case, the disk X which, for like reason, can readily project beyond X , serves as a support for the plane II , and the apparatus connected directly, and ensures that the angular motions of the plane II are made to depend therefore of the parts of any point of II , may be varied at pleasure by varying the the opposite direction, or it will be in motion, according as the mitre wheel is in gear with one or the other of the corresponding central wheels, and may rotate with each of them.

Detached altogether from the rest of the instrument, or what is better, supported on longitudinal pieces carried by the cross-pieces C, C we have a bridge (see fig. 1) capable of adjustment as regards height, and serving as a support for the pantograph- apparatus, or other connexion of the two plane II with the pencil which marks upon the other plane II .

It is hardly necessary to remark that in the simple form of the instrument where the disks X, X are themselves the planes II, II , then putting the mitre wheel of one plane A in gear with either of the corresponding central wheels, and making the other rotate, the other plane X will rotate as well; according as the mitre wheels are in the opposite direction, or it will be in motion, according as the mitre wheel is in gear with one or the other of the corresponding central wheels, and may rotate with each of them.

[510]

On Bicursal Curves.

[From the *Proceedings of the London Mathematical Society*, vol. IV. (1871–1873), pp. 347–352.
Read May 9, 1872.]

p. 181. A CURVE of deficiency 1 may be termed bicursal: thus it has a clear distinction according as the order is even or odd, and to fix the ideas I take it to be even.

A bicursal curve of the order n contains

$$\frac{1}{2}n(n+3) - \frac{1}{2}(n-1)(n-2) - 1, \quad = 3n \text{ constants;}$$

hence, if the order is $= 2n$, the number of constants is $= 6n$; such a curve is normally represented by a system of equations

$$(x, y, z) = (1, P)^n * (1, P^2)^n * y^2,$$

where Θ is a quartic function, which may be taken to be of the form $(1 - \theta^2)(1 - k^2\theta^2)$, or otherwise to depend on a single constant: viz. (x, y, z) are proportional to rational functions involving such a radical: since in the values of (x, y, z) we may consider the number of constants is $6(n+1) = (n-1) - 4 = 6n$, as it should be.

But the curve of the order $2n$ may be aberrantly or improperly represented by a system of equations

$$(x, y, z) = (1, P)^n * (1, P^2)^n * y^2,$$

viz. these equations, instead of representing a curve of the order $2n$ as said, provide a curve of the order $2n$, provided only there exist $2k$ common values of θ for which each of the three functions vanishes. And this represents a particular representation of (θ) of degree θ , with $z+1$, the consequent of θ which the

$$(x, y, z) = (1, P^n) * (1, P^2)^n;$$

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it is shown that such a transformation is possible, and a mode of effecting it, derived from a theorem of Hermite's in relation to the Jacobian H , Θ functions, is given in Clebsch's Memoir "Ueber Algegen Curven, deren Coordinates als rationale Funktionen eines Parameters und eines zweiten dort allgemeinen Funktion des selben dargestellt werden." *Crelle's Journal* (1864). Such a representation, in a very interesting way, will be made use of presently, but the more I am inclined to repeat the question: viz. given $\theta(p, q)$, $p^2q^2 = \theta(p, q)^2$.

Returning to the curves of deficiency 1, we see that a curve of the order $2n + 2$ contains θ each as a constants, and is normally represented by a system of equations

$$(x, y, z) = (1, P)^{n+1} * y^2.$$

Such a curve may be aberrantly represented; we may derive it by a rational transformation from the curve $(P = 1)$ of the order θ (θ being a positive integer); thus functions have then a common factor of the order θ , and omitting this, the equation is

$$(x, y, z) = (1, P)^{n+1} * y^2,$$

viz. the coordinates are functions of a quadriradiical equation; and which represents a system of (P, q) in terms of three by a system of equations

$$(x, y, z) = (1, P)^n y^2,$$

in the case of representation of the curve in terms of the form of which is the same as that of the curve $(P = 1)$ in the case of the form $(x, y, z) = 0$.

It is, however, to be observed that there are also curves of deficiency 1, similarly representable by a system of equations

$$(x, y, z) = (1, P)^n * y^2;$$

where the second factor is that belonging to the quadriradical determinant of system of (P, q) , where, A, B, C are quadratic functions of θ , viz. are then for example, $(1, P)^n y^2$.

In this particular, if $u = u$, then, instead of functions of the order $2n$, we shall find that, in fact, $u = (u, q)$, q proportional to functions of u , instead determines this view stress them in the form of $u = (u, q)^2$.

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$= B + \sqrt{\Omega}$ suppose; and then substituting in the equations $(x, y, z) = (1, P)^{n+1} * y^2$, we find

$$(x, y, z) = (1, \theta)^{n+1} * (1, P)^{n+1} y^2,$$

that is, here (x, y, z) , proportional to functions of u involving the quartic radical $\sqrt{\Omega}$; but these functions are of the order $u + 1$, not u .

In particular, if $u = u$, then, instead of functions of the order $2n$, we have functions of the order $u + 1$, $u + 1$, u ; that is, in this case the bicursal curve is properly represented in the form $(x, y, z) = (1, P)^{n+1} * (1, P)^{n+1} y^2$. This is no representation of an ordinary radical quadrant a may be exhibited by Cleach's Memoir.

We may, in fact, by linear transformations on u, v , reduce the quadripradiate relation to

$$\begin{aligned} 1 &+ u^2 \\ &+ 2hov \\ &+ v^2 \\ &+ cu^2v^2 \end{aligned} = 0;$$

that is,

$$1 + u^2 + 2hov + v^2 + cu^2v^2 = 0;$$

or putting herein $u + v = p$, $uv = q$, the relation is

$$1 + 2hq + p^2 - 2q + cq^2 = 0;$$

viz. extracting the square roots, this is

$$p^2 = -hq + 1 + 2(1 - 2h)q - cq^2;$$

$p^2 + bq = -1 + 2(1 - 2h)q - cq^2$; viz. extracting the square root, we obtain

$$\sqrt{p^2 + bq} = \sqrt{q} \sqrt{1 - 2(1 - 2h)q - cq^2};$$

if for shortness

$$Q = -1 + 2 - 2h(\theta) - cq^2;$$

we may then rationalise one of the radicals, for instance, by writing

$$-c\{-1 + 2 - 2h(c)q - cq^2\} = w^2 - 1 - c - q^2 + a; \text{ then, if}$$

$$op = -(1 - h)q^2 + b\sqrt{(1 - h)^2 + q^2} \left(q + \frac{1}{q} \right),$$

this becomes

$$\begin{aligned} -cq &= \{e - (1 - h)\} - \frac{1}{2} \left(q^2 + \frac{1}{q^2} \right); \\ &= \{e - (1 - h)\} - \frac{1}{2} \left(c^2 + \frac{1}{q^2} \right). \end{aligned}$$

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that is

$$\sqrt{q} = \sqrt{\left(\frac{(-1-h)c}{c}\right) \left(b + \frac{1}{q}\right)};$$

and the corresponding value of $\sqrt{p}$ is

$$\sqrt{p} = \sqrt{\left[\frac{1-1-h}{2} \frac{c}{c} \left(b + \frac{2}{c}\right) - h^2\right]},$$

where q stands for the value

$$\frac{1}{2} \left[1 - h + \sqrt{1 - h^2} \left(c + \frac{1}{c}\right) \right].$$

We may write them in the form

$$1 : \sqrt{Q} : \sqrt{q} \cdot \sqrt{Q'} = M : 1 + cP^2 : \sqrt{q^2},$$

where M is a constant, and Θ is a quartic function of θ , such that $(1 + cP^2) - \Theta$ is a quadric function only of θ .

The equations

$$(x, y, z) = (1, P)^{n+1} * y^2,$$

then assume the form

$$(x, y, z) = (1, P^{n+1} * (1, P)^{n+1} * \sqrt{Q^{n+1}};$$

and on the right-hand side the term of the highest order in θ is

$$(1 + cP^2)^{n+1} \theta^{n+1};$$

viz. if $n = cP^2 = 0$, then this is

$$(1 + (1 - cP^2)^{n+1}) \theta^{n+1} = \sqrt{q^{2n+1}};$$

which is the same as the term of the highest order in θ in

$$(x, y, z) = (1, P^{n+1} * \sqrt{q^{2n+1}};$$

In particular, if $n = u$, then, instead of $2n + 1 = 2u + 1$, $u + 1$, in the case of the order $2u$, as supposed at first, we have $u + 1$, $u + 1$; but in this form, expressing u in terms of u , the equation are

$$(x, y, z) = (1, P^{n+1} y^2, (*P^{n+1} y^2),$$

viz. (x, y, z) are connected by a quadripradiate equation, is also represented by the equations

$$(x, y, z) = (1, P^{n+1} * y^2, (*P^{n+1} y^2),$$

which is the required transformation of the original equations.

It is to be noticed that the foregoing form

$$1 + u^2 + 2hov + v^2 + cu^2v^2 = 0$$

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is the most special form to which the quadriradiate relation can be reduced by the linear transformation of u, v ; in fact, by mere division, the equation is made to have the constant term 1; by mere replacement of variables, the equation in u takes the form $u^2 = uv + cv$; and as before, by means of u , we may, by a linear substitution, reduce it to this form, viz. we may write

$$\theta = P + \theta_1(1 - x_1t) - F(1, c)P^2;$$

which is of the form general that if we had

$$(1, a)^{n+1} * y^2P + y^2\alpha;$$

viz. there is a hyperboloeous constant θ . And we then see how it is that the system of equations

$$(x, y, z) = (1, P^n * y^2);$$

(a, b connected by a quadrirat equation, contain c and $b + 1$ constants, and only of these being numerical.

Clebsch's transformations, above referred to, is as follows:

Starting with the equations

$$(x, y, z) = (1, P)^{n+1} * (1, P)^{n+1} * \sqrt{y^2(\theta)};$$

if the function Θ is not originally of the standard form, we may, by a linear substitution, reduce it to this form, viz. we may write

$$\Theta = F(\theta) - F_1(\theta) - F_1(\theta);$$

and then writing $\theta = \alpha t$, ($u(\theta = \alpha t)$), we have

$$\sqrt{\Theta} = \alpha t \cdot (-u\alpha t) - \dots = au\alpha = u\alpha + \dots;$$

so that the formulas become

$$(x, y, z) = (1, P)^{n+1} * (1, P)^{n+1} * \sqrt{\Theta^{n+1}};$$

Hermite's theorem, used in the demonstration, is that any such function of u is expressible in the form

$$\xi \cdot \frac{H(u - a_1)H(u - a_2) \cdots H(u - a_n)}{H(u)^n},$$

where $\xi, a_1, \dots, a_n$ are constants, $a_1 + a_2 + \dots + a_n = u$.

(If, θ denoting here the two Jacobian functions); where

$$\xi_1 + \xi_2 + \dots + \xi_n = u.$$

Observe, in passing, that the equation

$$\Theta = (1 - u^2)(1 - x_1u^2)P, u + cu^2 + \dots + cu^2 = uP + \dots = 0.$$

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gives, when rationalised, an equation in ξ 's of the order $2n = 2n$; this form of this equation are $u, u, u, x_1 + x_2$, &c. Considering the functions $(1, u^2)^{(2n)}$ and $(1, u^2)^{(n+1)}$ as indeterminates, the coefficients can be found so that all the rows of the roots of the equations is u^2 shall have any given values whatever; $u^2a, u, u^2 = u^2b$, &c.; the theorem then shows that the values of the remaining root is $u^2a + b + c$, where

$$-x_{a_1} = u_1, u_2, \dots, u_n, \quad -x_{a_2} = -(a_1a_2 + \dots + a_n)$$

which is, in fact, Abel's theorem.

Now, supposing that the three functions of θ vanish for 23 common values of θ , and the functions of u will continue at the most $u + 1$. From this, the equation are then

$$(x, y, z) = (1, w)^{n+1} * y^2;$$

we have the set of equations

$$(x, y, z) = (1, P)^{n+1} * \frac{F(c + a_1) H(c - a_1) \dots H(c - a_n)}{H(c)^{n+1}};$$

where, however,

$$a_1 + a_2 + \dots + a_n = u.$$

[the values $a_1, \dots, a_n$ are course different for the three coordinates x, y, z respectively; viz. we have

$$\begin{aligned} a_{x_1} + a_{x_2} + \dots + a_{x_n} &= u, & -\dots &= (a_{x_1} + a_{x_2} + \dots + a_{x_n}) = u, \\ a_{y_1} + a_{y_2} + \dots + a_{y_n} &= u, & -\dots &= (a_{y_1} + a_{y_2} + \dots + a_{y_n}) = u, \\ a_{z_1} + a_{z_2} + \dots + a_{z_n} &= u, & -\dots &= (a_{z_1} + a_{z_2} + \dots + a_{z_n}) = u, \end{aligned}$$

Writing the case

$$u = a + u', \quad u_1 = u_1 + u', \quad u_2 = u_2 + u', \quad \dots, \quad u_n = u_n + u',$$

and consequently

$$u'_1 + u'_2 + \dots + u'_n = 0,$$

or, changing the common denominator,

$$(x, y, z) = \frac{H(u - u'_1)H(u - u'_2) \dots H(u - u'_n)}{H(u)^{n+1}};$$

where

$$u'_1 + u'_2 + \dots + u'_n = 0;$$

or, what is the same thing,

$$(x, y, z) = (1, u' + u^*)P, \quad (1, u + a^*P^*)y \dots u_n;$$

viz. writing $u_1, u' = c$, and $u_2 (1 - c)(1 - kc)$, we have a system of the form

$$(x, y, z) = P, \quad PP^{n+1}, \quad y \dots y \dots;$$

a normal representation of the curve of the order $2n$.

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The relation between the parameters θ, θ' is given by $\theta'^2 = m^2(u - x)$, $u'^2 = \theta$, that is, we have

$$e^{\sqrt{\theta}} = \frac{\sqrt{\theta}x_1 du + a_1 \sqrt{(1 - x_1^2)(1 - 2\theta_1)}}{1 - 2u + x_1 \cdot \theta},$$

or, writing $uv/u = u$, this is

$$\theta' = \frac{c\sqrt{x_1} + b_1 \sqrt{(1 - c^2)(1 - kc^2)}}{1 - x_1 b^2};$$

and, conversely, we have

$$\theta' = \frac{c\sqrt{x_1}(1 - x_1) - kc) + a_1 c \sqrt{(1 - c^2)(1 - kc^2)}}{1 - x_1 c^2};$$

and the theorem, in fact, shows that, substituting this value of θ in the functions of θ which serve to express x, y, z , these become proportional to the functions of θ' ; viz. they become equal to these functions, each multiplied by an irrational function $A^k + B^k a \sqrt{\theta}$ (A, B rational functions of θ).

[511]

Addition to the Memoir on Geodesic Lines, in particular those of a Quadric Surface (508).

*[From the Proceedings of the London Mathematical Society, vol. IV. (1871–1873), pp. 268–286.
Read June 13, 1873.]*

p. 188. 38. In the Memoir above referred to, speaking of the geodesic lines on the above hyperboloid, I say (No. 35), “The geodesic of initial direction $M1$ touches at M the oval curve of curvature M_2 , and lies wholly above this curve; it makes an infinity of convolutions round the upper part of the hyperboloid, cutting all the oval curves of curvature through M , viz. it cuts all the hyperbolic curves the parameters of which are between $-b$ and $-a$; and so on,” but in fact such convolutions are incorrect; I was led to it by the assumption that the geodesic could not touch any hyperbolic curve of curvature, but in fact, looking at the geometry the geodesic actually cuts the hyperboloid being defined: such hyperbolic curve of curvature touched by the geodesic, &c.; that is, in the wholly part of the geodesic the value $b+c$ may be taken arbitrarily; while $b+c$ may be a positive value $b+c$ may be a negative value $b+c$, which is may be a positive value below $b+c$ may be then negative, that the term of integral consideration obviously enough, that $b+c$ value of $b+c$ has the contrary; viz. it but it is greatly preferable to bring the analytical considerations to bear with the matter in fact the questions to ratifying $b+c$; viz. if even $b+c$, a value $b+c$ ratifying $b+c$, a value $b+c$ ratifying $b+c$, a value $b+c$.

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39. I have effected, for a particular skew hyperboloid and oval curve of curvature thereof, the numerical calculations for laying down the geodesic lines which touch the curve of curvature. Taking in general the equation of the hyperboloid to be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1,$$

and the b -curves of curvature to be the intersection by the cylinder

$$\frac{x^2}{a^2 + \theta} + \frac{y^2}{b^2 + \theta} = 1;$$

then the satisfied values for the hyperboloid, and oval curve of curvature touched by the geodesics, are

$$\begin{aligned} a &= 900, \\ b &= 400, \\ c &= -300, \\ \theta &= -1600, \end{aligned}$$

so that a, b, θ are the positive values 900, 400, 300, 1600, and 1650 respectively. I would that $p = a$ is given the oval curve of curvature, viz. $p = -\theta$, the elliptic principal section; $p = b$, the green oval curve, and that we are in the manual concerned only with the oval curves above this, for which p lies between $-c$ and $-\theta$. We are in the manual to deal with the integrals

$$\Pi(p) = \int_b^p \frac{dp}{\sqrt{(a+p)(b+p)(c-p)(p-\theta)}},$$

and

$$\Psi(p) = \int_b^p \frac{dp}{\sqrt{(a+p)(b+p)(c+p)(p-\theta)(p+c)}},$$

or if $p = c = c$, extending from 0 onwards, viz.

$$\Pi(p) = \int_0^p \frac{dp}{\sqrt{(a+b+p)(c-p)(p+c)(p+c)}};$$

the relation for any particular position of the point P and P' .

40. To avoid discontinuity as to sign, it is convenient to take the integral $\Psi(p)$ in a particular manner. The hyperboloid is by the α and perpendicular plane divided into four quadrants; we may attend only to the upper half of the hyperboloid, say this upper half is then divided into four quadrants, viz. $p > 0, y > 0, y < 0, x > 0, x < 0$. In the first quadrant the geodesic curves are one and the same curve; if we let $y < y$, &c.; that is, as the hyperboloid extends referred to in the same fig. 1, we may consider the quadrants as forming an infinite succession round, half, third, &c., not, &c., (478).

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and so on; or we may take them in the reverse order, $-1, -2, -3$, &c. For a hyperbolic curve $q = -b - x$ in the first quadrant the integral is to be taken $x = 0$ to $x = c$; for a curve in the second quadrant $x = 0$ to $a - b$, thence $a - b$ to 0 , and thence 0 to π ; and so on; and so for a quadrant in the quadrant -1 , the integral is to be taken from 0 to π , taken negatively, &c.; that is, as the hyperbolic curve $b + b$ travels from the positive direction, $b - c$ the integral $\Psi(b)$ continually increases from zero; and if the curve travels from the b -direction, $b - c$ the integral $\Psi(b)$ continually decreases from zero; that is, it increases negatively. It is to be remarked that this integral $\Pi(b)$ requires no calculation; the subsequent value $\Pi(c + c)$ value of b values $b + c$ to b values $b + c$.

The integral $\Pi(b)$ requires no explanation; it is taken from $u = 0$, giving a certain oval curve, up to any positive value of u , giving the oval curves above this one; and, in particular, taking the integral to $u = \infty$ (or, what is the same thing, to $p = \infty$) its value is finite, $= K$, suppose.

41. Consider the geodesic which touches the given oval curve at a point for which $\Psi(p)$ has a given value Q ; at the point $p = \theta$, the integral $\Psi(p) = 0$; we may write the equation of the geodesic $\Psi(p) + \Pi(p) = C$, and consequently

$$\Psi(p) = Q + \Pi(p).$$

Taking the positive sign, then as p increases from θ , $\Psi(p)$ increases, or the describing point of the geodesic moves upwards from the point of contact in the direction of positive rotation; and taking the negative sign, then $\Psi(p)$ decreases, or the describing point of the geodesic moves upwards from the point of contact in the direction of negative rotation; and, in particular, p becoming infinite, then on the upper half of the hyperboloid, the oval curve, for which $\Psi(p) = K$, will be reached, the second branch that for which $\Psi(p) = -K$.

42. The graphical process is as follows: we describe, on the hyperboloid, a series of hyperbolic curves of curvature, numbering them according to the values of $\Psi(p)$; viz. considering the hyperbolic branches which form the x, y, z &c. and $y \pm$ sections respectively, these are $0, 2000, 4000, \dots, 34726$; and on going round a second time they would be $K + 34726, K + 36000$, &c. respectively, and so on; the same dispersion of the direction of negative rotation, numbering them according to the values of $\Pi(p) = K - Q + \Pi(p)$, &c. ($p = K$), where the oval curves of curvature is the ellipse which is a principal section of the hyperboloid, and then even to attain to it by value 1 except for an oval curve of curvature.

In the example, and showing belonging thereto (1), where, for convenience, the values of the integrals have been multiplied by $100,000$, we have, as an upper limit, $K = 34726$; the two sets of curves are drawn at intervals of 2000 ; viz. we have the hyperbolic curves $0, 2000, 4000, \dots, 34000, 34726$; and the oval curves $0, 2000, 4000, 6000, 8000, 10000$; the distance of the successive oval curves increases very rapidly, since the curve at infinity would be $K, = 12400$, and the curve $K = 10000$ is the last which comes into the limits of the figure.

[This drawing was exhibited at the meeting of the Society.]

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43. The two sets of curves of curvature being thus drawn at equal intervals of $\Pi(p)$ and $\Psi(p)$ respectively, drawing the hyperboloid into quadrilateral spaces (which of course should theoretically be indefinitely small), the diagonals of these quadrilateral spaces are the elements of the geodesic lines: that is, starting with the geodesic of equation $\Psi(p) = C$, and following its identity, we may readily trace it.

In the case in question, the data are as follows:

$$\begin{aligned} a &= 1600, \\ b &= 1600, \\ c &= 1000, \\ b - c &= 30 = 1000; \\ \theta &= -1600; \end{aligned}$$

where putting in $p = \theta$, θ is found, the equation

$$J = \int_b^p \frac{dp}{\sqrt{(a+p)(b-p)(c+p)(p-\theta)}}.$$

where J , the integral calculated up to a somewhat large value $p = p_2$.

Writing

$$\begin{aligned} u &= \frac{1}{2}(a + 2p), \\ \theta &= \frac{1}{2}(b - 2p), \\ \eta &= \frac{1}{2}(c + p) - b, \\ b &= -\frac{1}{2}b - 2c) - b, \end{aligned}$$

where as said a are at yet undetermined, we have

$$\begin{aligned} (p+a)(p+b)(p+c)(p+a) &= u^2 - b^2, \\ (p-a)(p-b)(p-c)(p-a) &= u^2 + b^2; \quad b > 0 = b^2, \\ (p-a)(p-b)(p-c)(p-a) &= u^2 - b^2, \\ (p-a)(p-b)(p-c)(p-a) &= u^2 + b^2; \quad b < 0 = b^2, \end{aligned}$$

and the integral is then

$$J = \frac{1}{2} \int_b^p \frac{dx}{\sqrt{(b-a)(b+b)}}.$$

But we may determine in a such that for $p = a > p$, $(p-a)(p-b)(p-c) > a(p-b)$, that is, $(p-a)^2 + b^2 > 0$, and the integral is then

$$J = \frac{1}{2} \int_b^p \frac{dx}{\sqrt{(b-a)(b+b)}}.$$

The determination then depends on the two roots of the mentioned integrals, the values of which are calculable by writing therein $\sqrt{q} = u$, viz. we have

$$\int \frac{dx}{\sqrt{q+b(c+b)}} = \frac{1}{c-b} \log \frac{(c-b)x^2 + 2cx + b}{c-b}.$$

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and hence, substituting in the formula, for L, c its value

$$\frac{2 \log a}{c - b} - 1,$$

the superior limit is

$$L_1 = \frac{1}{c - b} \left\{ \sqrt{q} \log \frac{\sqrt{(p+a)(p-b)} + \sqrt{(p+a)(p+a)}}{\sqrt{p^2 - a^2}} \right\};$$

and the inferior limit is

$$L_2 = \frac{1}{c - b} \left\{ \sqrt{q} \log \frac{\sqrt{(p+a)(p-b)} + \sqrt{(p+a)(p+a)}}{\sqrt{p^2 - a^2}} \right\};$$

where denom. = $(a - p)b$, and these resulting values of L, L_2 , are obtained by writing therein $b + p$ in the place of b .

45. The numerical values are $p = 19800$; $a, b, b', c = 800, a, b, c = 1605, 1745$; and thence determining by trial values of a and b ,

$$\begin{aligned} a &= 600, \\ b &= -500, \\ a' &= 1480, \\ b' &= 145, \\ \beta &= 1825, \\ b' &= 1825 + 165 = 1825. \end{aligned}$$

I obtained the logarithmic and circular terms of the two limits respectively;

	Superior	Inferior
Logarithmic	015668	015144
Circular	002522	005003
	023720	020737.

The value of L was $10411 \times 100000 = 104111$; and the two limits then are 124390 and 123450; or removing the factor $\frac{1}{100,000}$, they are 1.2439 and 1.2345; the mean of which 1.2400, was taken for the value $\Pi(p), p = \infty$; that is $K = 12400$.

46. As regards the calculations of the integrals $\Pi(p)$ and $\Psi(p)$, introducing the numerical values, and multiplying by the before-mentioned factor 100,000, we have $\Pi(p) = 400$

$$\Psi(p) = 100,000 \int_0^u \frac{du}{\sqrt{(1500 - u)(150 + u)(2000)(u + 2050)}},$$

which for any small value of u is

$$= 100,000 \frac{2u}{(150 \cdot 100 \cdot 2000 \cdot 2050)^{1/2}} \left(\int_0^u \frac{du}{\sqrt{\dots}} \right) = 2\sqrt{u};$$

viz. this is

$$= 88\sqrt{2} \log = 27001895\sqrt{u},$$

which was used for the values $u = 1, 2, \dots, 10$, that is, to $q = -610$. When the calculation was continued by quadrature systems of u at values $u = -1, -10, 31, 30, \dots$, up to $u = -10000$, $q = -900$. The remainder of the integral, setting from $1000 = u = w$ (w at $u = w$ etc.) was here continued.

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which was used for the values $u = 9, 8, 7, \dots, 1, 0$, to complete the calculation up to $q = -300$.
47. We have in like manner, $p = 1650 + u$,

$$\Pi(p) = 100,000 \int_0^u du \sqrt{\frac{1600}{2500(u + 1650)(u + 2050)(u + 50)}}$$

which for small values of u is

$$= 100,000 \cdot \frac{1}{\sqrt{2500 \cdot 1650 \cdot 2050 \cdot 50}} \int_0^u du = 5\sqrt{u},$$

viz. this is

$$= 5825 \log 5 = 270111035\sqrt{u},$$

used for $u = 1, 2, \dots, 10$, that is to $p = 1600$. The calculation was afterwards continued by quadrature, by giving to u successive integer values, $u = 10, 20, \dots, 100, 200$, &c., as far as the limit; wherefore, as we approaching above, the value for $p = -\infty$ was found to be $= 12400$.

48. After the calculation of the values of $\Pi(p)$ and $\Psi(p)$, it was easy by interpolation to revert these tables, so to obtain a table which, for if to Ψ as argument, gives the values of p (q), as the limits of the equations are real of the geodesic lines, $b + (a + c, c, e)$, $\Pi(p) = 12400$ is which the analyte agreeing with this; we are also enabled, by an interpolation of one of the calculation in the form

$$\Pi(p) = (q - 400)A + (q - 400)^2B + (q - 400)^3C,$$

to take from this table the equations of the geodesic line as referring in the form $\Pi(q) = (q - 400)A + (q - 400)^2B + (q - 400)^3C$, &c.; viz. to take the geodesic line $\Pi(q) = (q - 400)A + (q - 400)^2B + (q - 400)^3C$, &c.; &c.; the same that the same form of the curve, and by taking the same value $\Psi(p) = (q - 400)A + (q - 400)^2B + (q - 400)^3C$ on the same form ($p = \text{const.}$), we ensure that the curve is sufficiently well to the curve. This curve, for $p = 0$, was readily found to be

$$\Pi(p) = -1.0 \cdot \Pi(q).$$

But it is possible to derive a more elegant form for $p = 0$ from the equation

$$\Psi(0) = 12400 \cdot Q + Q \cdot \Pi(0).$$

Observe that the integral $K' = 34726$ in the case considered measures the quadrant of the hyperboloid, viz. to value $\Psi(p) = K'$ determines two hyperbolic curves of curvature (principal sections), the natural distance, that is, an ellipse, &c. which is included in any $p = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K'$ at $\Psi(p) = K' = 34726$ at hyperbolic, including $p = \text{const.} = K'$. Observe that the form of these foregoing equations gives one geodesic in the manual containing $\Psi(p) = K' + Q$, we have the geodesic, in measured by the value Q , $\Pi(p) = K' (p = K')$.

50. Now it is easy to see that on the oval curve of curvature appearing the principal elliptic section, that is, as p approaches 0 (or nothing) $\Psi(p) = -\infty$, as on

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diminution towards zero; the integral K' shows its value only slowly, increasing towards a certain constant limit; but, contrariwise, K increases without limit, its value at any small value of u being of the form $A - B \log p$, in the limit, when $K = \infty$, A diminishes, the value of $\frac{dp}{dQ}$, the amplitude of the geodesic, continually diminishes; that is to say, the geodesic continues, at it makes round in any number of convolutions, more and more nearly to the form of the principal section of the geodesic at all events convergent towards a special, value at every approximate convolutions b , $b \cdot c$, that is the same as the principal section. But it is possible to derive some general numerical results from the analytical formulas.

51. To sustain the foregoing statements, I write $\theta' = (-c + a) + 1000 \cdot s$, and I consider the integral

$$K_a = 100000 \int_0^a \frac{ds}{\sqrt{(s+1)(150+s)(150+100)(2000)(s)}},$$

say for a moment this is

$$K_a = \int_0^a U_a ds.$$

Supposing a to be small, we divide this integral into two parts, say from 0 to θ (θ where $a, a+c$ to comparison with a , and so on; in comparison with a , but small in comparison with $a, 1000, 9$ to a , where the second part the expression under the integral sign and the value of the integral varies slowly with θ , and we may, as an approximation, write $s = a$. We have then

$$K_a = \int_0^a U_a ds + \int_0^a U_a ds,$$

and the first part listed to be written

$$K_a = 100000 \int_0^a \frac{ds}{\sqrt{(1500 \cdot 100 \cdot 2000)(2050)}},$$

viz. the integral is here

$$A \cdot \int_0^a \sqrt{s(2050+a)} ds = A \log \frac{a + \sqrt{(2050+a)}}{\sqrt{2050}} = A \log \frac{4a}{2050} \text{ approximately,}$$

or say this is

$$= \log \frac{4a}{2050};$$

The first term is thus

$$= 100000 \int_0^a U_a ds = 100000 \int_0^a U_a ds \int_0^{a+c} U_a ds \int_0^{a+c} U_a ds,$$

which is

$$= 4419 \log \frac{4a}{2050}.$$

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or we have

$$K_a^* = 4119 \log \frac{4a}{2050} + \int_0^a U_a ds.$$

We ought to have the same value if the integral, whatever, within proper limits, the assumed value of a may be. Taking, for instance, $a = 10$, and $a = 100$, we ought to have

$$\begin{aligned} K_a^* &= 4119 \log \frac{400}{2050} + \int_0^1 U_a ds \\ &= 4119 \log \frac{400}{2050} + \int_0^1 U_a ds. \end{aligned}$$

that is,

$$4119 \log 2 = \int_0^1 U_a ds.$$

In verification, I calculated the second side by quadrature; viz. for the values 30, 60, 70, 80, 90, 100, the values of U_a are 30.532, 25.711, 23.812, 22.527, 17.604; whence, adding the half ends and the extreme terms to the sum of the mean terms, and multiplying by 10, the value of the integral is = 13354. The value of the left-hand side is = 11873, which is a sufficient agreement.

52. Returning to the formula for K_a^* , this may be written

$$K_a^* = (4119 \log 400 + \int_0^a U_a ds) = 4119 \log a;$$

I did not calculate the value of the integral in this formula, but determined the term in () to such value that the formula should be correct for the foregoing value $a = 50$, viz. the term is then

$$= 12400 + 4119 \log 50 = 12400 + 4598 = 19488, \text{ or say } 19500;$$

we thus have

$$K_a^* = 34726, \text{ or say } 19500.$$

We thus see how to give to so such a value that the quantity $\frac{K}{2k}$, which is the number of convolutions of the geodesic, may have any given value: and, in particular, we see how exceedingly slowly such an number of convolutions is, even at the small value of a which corresponds to a geodesic which is most by any moderately large number of convolutions; for instance, $a =$.

CONCLUSION. Instead of moving an above, of a geodesic at touching of an arbitrary curve, a hyperbolic curve of curvature, the accurate expression is that the geodesic at infinity is parallel to a certain hyperbolic curve of curvature. The geodesic has, in fact, for its asymptote in the asymptote $b + b$ a $b + b$ a $b + b$ a $b + b$ a $b + b$ a $b + b$ a $b + b$. Added Dec. 1873.

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Tables I and II: numerical values of $\Pi(p)$ and $\Psi(q)$.

TABLE I.		TABLE II.	
p	$\Pi(p)$	q	$\Psi(q)$
1650	000	−400	000
1	502	1	843
2	711	2	1249
3	870	3	1550
4	1003	4	1747
5	1124	5	1975
6	1230	6	2184
7	1329	7	2337
8	1421	8	2493
9	1507	9	2620
1660	1589	410	2734
70	2186	20	4016
80	2603	30	4860
90	2933	40	5782
1700	3208	50	6402
10	3449	60	6874
20	3642	70	7304
30	3823	80	7711
40	3974	90	8050
50	4112	500	8385
60	4239	600	11288
70	4351	700	13580
80	4453	800	15470
1800	4642	1000	18482

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TABLES I, II, III (CONTINUED): cf. book pages 197–199.

p. 198–199. Numerical Tables III and IV (continued).

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On a Correspondence of Points in Relation to Two Tetrahedra.

[From the *Proceedings of the London Mathematical Society*, vol. IV. (1871–1873), pp. 396–404.
Read June 12, 1873.]

p. 200. THE following question has been considered by E. Sturm in an interesting paper, “Das Problem der Projectivität und seiner Anwendung auf die Flächen zweiten Grades,” *Math. Ann.*, t. I. (1869), pp. 333–374: Given in a plane two groups of the same number (5, 6, or 7) of points, to find points P , P' homographically related to these two groups respectively; viz. the lines from P to the points of the first group and those from P' to the points of the second group are projectively equal. The solution depends on the resolution of the equation systems by the rules of the ten tetrahedra; it is required to find the locus of the points P , P' which are homographically related to two such groups of A, B, C, D and A', B', C', D' wherein in line, and to find the locus of P , P' .

If the points A, B, C , or P and the points A', B', C' , or P' are subtended equal angles, then we may say all of these tetrahedra $ABCD$ and $A'B'CD'$; the four cubic surfaces, four of each in space, will pass through the points A, B, C, D and the four tetrahedra of the four cubic surfaces $ABCD, A'BCD, A'B'CD, A'B'C'D$, &c. that the time of the cubic surfaces $A'B'C'D', A'B'C'D', A'B'C'D'$, &c.; and the like for the locus of P' .

If the points A, B, C, D and P' and the points A', B', C', D' subtended equal cosines, the line from each of the points A, B, C, D to the locus point of infinity $ABCD, A'BCD, A'B'CD'$; and the like for the locus of P' .

If the points A, B, C, D and P and the points A', B', C', D' and P' subtended equal angles, then the two tetrahedra are correspondingly homotopic, viz. the locus of P subtends an isolated angle which determined as a as homotopic ($ABCD, A'B'CD, A'B'C'D$), and the like for the locus of P' .

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Finally (although this is a theorem which I do not require), if the points A, B, C, D, E and P and the points A', B', C', D', E' and P' subtended equal angles, then there are three positions of each point.

The problem I propose to consider is: Given the tetrahedra $ABCD$ and $A'B'CD$, it is required in the plane ABC and $A'B'C'$ respectively to find the points P, P' such that A, B, C, D and P , and A', B', C', D' and P' subtended equal angles. I was led to this by the more general problem, which I am at one moment proceeds: viz. the points of a configuration of four planes; intersection of these triple planes which determined by the parameters A, B, C, D of the four points A, B, C, D are at A, B, C, D (A, B, C, D); the four edges of the tetrahedron $ABCD$, etc. at six relative six relative to the points of the four planes, namely

1. P is such that A, B, C, D and P , and A', B', C', D' and P' subtended equal angles; locus a cubic line through $A, B, C, D, (ABC), (ABE), (ACE), (BCE)$.
2. P is such that A, B, C, D and P , and A', B', C', D' and P' subtended equal angles, and that PE and PE' are in a given line; locus a certain ruled curve;

and the required position of P are obtained as the intersections of the two loci.

I proceed to the analytical investigation.

Preliminary Formulæ.

1. Consider a triangle ABC , and let the position of a point P be determined by means of its coordinates x, y, z , which are equal to the perpendicular distances of P from the sides, each divided by the perpendicular distance of the opposite vertex (as usual; x, y, z are positive for a point within the triangle), or what is the same thing, $x, y, z = PBC, PCA, PAB$, divided each by ABC ; whence identically $x + y + z = 1$.

Suppose for a moment the rectangular coordinates of A, B, C are $(a_1, b_1), (a_2, b_2), (a_3, b_3)$ respectively; and those of P are X, Y . Also let the rates BC, CA, AB be $= a, b, c$ respectively.

We have

$$X = a_1x + a_2y + a_3z,$$

$$Y = b_1x + b_2y + b_3z,$$

$$1 = x + y + z;$$

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and if we consider a second point P' , the coordinates of which are x', y', z' and X', Y' , we have the like relations between these quantities. Calling b the distance of the two points P, P' , we may write

$$\begin{aligned} D &= [a_1(x - x') + a_2(y - y') + a_3(z - z')]^2 \\ &\quad + [b_1(x - x') + b_2(y - y') + b_3(z - z')]^2 \\ &= [(b_1 - b_2)(z + y) - (b_3 - b_2)(z - y') - (b_3 - b_1)(z + z')]^2 \\ &\quad + [(a_1 - a_2)(x + y) - (a_3 - a_2)(x - y') - (a_3 - a_1)(x + x')]^2; \end{aligned}$$

the last term being in fact -0 ; viz. this is

$$\begin{aligned} D &= (a^2(y - y') + (b^2(x - x'))(z - z') + (a^2(x - x') + b^2(y - y'))(z - z')) \\ &= [\beta(x - x') + y(z - z') - z(y - y')]^2 + (\beta(x - x') + z(y - y') - x(y - y'))^2, \end{aligned}$$

or what is the same thing, the coordinates for the distance D of the two points P, P' which expression may be modified by means of the identical equations

$$1 + x + y + z = 1, \quad 1 + x + y + z = 1.$$

[full quadratic forms continued]

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3. Consider next a tetrahedron $ABCD$, let the position of a point be determined by 4 coordinates x, y, z, w relating to the faces (the sum of these being $= 1$); and let the four points A, B, C, D corresponding to these coordinates have the coordinates of unit area as above. By means of the same construction we now express the distance of two points P as a function of the coordinates of P .

In fact for the tetrahedron $ABCD$ and the point P , we have a system of relations $(A, B, C, D)(x, y, z, w)(X, Y, Z, W)$ as before, viz. taking A, B, C, D for the angles, and similarly those of vertical triangles (B, C, D) , (C, D, A) , (D, A, B) respectively. Here we require for the case the perpendicular and ratios of the tetrahedron $ABCD$, X which enter into these manuscripts equations.

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viz. this is the same thing,

$$\Sigma = a^2yz + b^2zx + c^2xy + a^2dz(z + y - z),$$

Moreover, if Δ a value also are an area of the triangle, then

$$(2\beta_1 - \beta_2)\Sigma_2 \cdot V \cdot s_1 \cdot \zeta_4 - s_4 + \zeta_2 \cdot (s_1 + s_2 + y + s) - \beta_1 \sin A_1(s_1 + s_2 + y + s) = 0;$$

so that the equation becomes

$$-(2\beta_1 - \beta_2)\Sigma_1(c + (b_1 - \zeta_3))^2(c_3 - c_2)(x + y + 1) + \beta(c_3 - c_2 + 2\beta_2)(x + y + 1)(c_3 - c_2 - 2c)(c_2 - c_3) - 2c\beta_3(2 + (c_1 - 1)) - 2c\beta_3 = 0;$$

so that this represents the like equation

$$-c^2y(x_1 + b)y^2 + b\beta(c_1 - 1) + b\beta b_3(c_2 - c_1) = 0,$$

or forming the like equations of the other similar circles, we have the circles (B, C) , (C, A) , (A, B) containing the angles E , E , F respectively; and the equations are

$$U + \Delta(c\zeta - \cot A)(c\zeta - 0,$$

$$U + \Delta(\zeta + \cot B)(\zeta = 0,$$

$$U + \Delta(\zeta - \cot C)\zeta = 0.$$

Correspondence, A, B, C, D and A', B', C', D' subtending equal angles.

5. Consider now the two figures A, B, C, D, P and A', B', C', D', P' subtending all the same angles E, F, G, H enclosed at P, P' , then we have

$$\frac{U}{ABCD} = \cot E = \cot A, \quad \frac{U'}{A'B'C'D'} = \cot E = \cot A',$$

$$\frac{U}{ABCD} = \cot E = \cot B, \quad \frac{U'}{A'B'C'D'} = \cot E = \cot B',$$

$$\frac{U}{ABCD} = \cot E = \cot C, \quad \frac{U'}{A'B'C'D'} = \cot E = \cot C',$$

and thence

$$\frac{U}{ABCD} + \cot A = \cot A',$$

$$\frac{U}{ABCD} + \cot B = \cot B',$$

$$\frac{U}{ABCD} + \cot C = \cot C'.$$

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and consequently

$$\begin{aligned}\frac{1}{x'} \cdot \frac{1}{y'} \cdot \frac{1}{z'} &= \frac{U'}{ABCD} \cot A = \cot A', \\ \frac{1}{ABCD} \cot B &= \cot B', \\ \frac{1}{ABCD} \cot C &= \cot C',\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}x' : y' : z' &= \frac{x}{U + \Delta(\cot A - \cot A') V_3} \cdots ; \\ \overline{U + \Delta(\cot B - \cot B') V_2} &\cdots ; \\ \overline{U + \Delta(\cot C - \cot C') V_3} &\cdots ;\end{aligned}$$

where observe that the equations

$$\begin{aligned}U + \Delta(\cot A - \cot A') V_3 &= 0, \\ U + \Delta(\cot B - \cot B') V_2 &= 0, \\ U + \Delta(\cot C - \cot C') V_3 &= 0,\end{aligned}$$

represent circles (B, C) , (C, A) , (A, B) containing the angles A' , B' , C' , and since $A' + B' + C' = \pi$, we have these three circles meeting at a point. We may for convenience write

$$x' : y' : z' = BCO : CAO : ABO,$$

suppose, X, Y, Z are quartic functions of x, y, z , and the curve is the first one representing a X, Y, Z of the order 4; viz. since the curves (BC) , (BC) , &c., is an algorithm which will be common point through A, B, C , of the first figure—corresponding to the locus on P' , of the second figure—the point at infinity $X = X' = Y' = Z' = 0$.

We have therefore

$$x' : y' : z' = \frac{x}{BCO} : \frac{x}{CAO} : \frac{x}{BCO},$$

viz. this is a quartic curve having the points A, B, C on the line BCO and ABC , the expressions being multiplied into the proper constant factors to make the constant factors whereby x, y , and so a, b, c of x, y, z identically as $A.BC.OA.B.OD.OA$. identically $Y = X = (A.B.D.A)$, and similarly $Y = (A.B.D.G.)$.

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Correspondence, A, B, C, D and P and A', B', C', D' subtending equal angles.

6. Consider now in plane the points A, B, C, D which on P , and the points A', B', C', D' which on P' , subtend equal angles. Let a, b, c, g, h, k denote the perpendicular distances of P from the lines BC, CD, AD, AB, AC, BD respectively; and let the lines of P, P' be denoted by A, B, C, D, G, H, K where we have G, H, K denote the corresponding angles in the two figures: then in respect to P and the four points A, B, C, D , we have three equations of the above form, viz. where the value $\frac{1}{x'}$ corresponds to the equation in the first figure;

$$a' : g' : k' = BCD'O : CDA'O : DAB'O,$$

the expressions being multiplied into give proper constant factors to make the constant terms whereby x, y, z of a, g, h of a, g, h of a, g, h of a, g, h of a, g, h as before, a, b, c, g, h, k , respectively.

We have in like manner

$$a' : b' : c' = \frac{a}{BCD'O} : \frac{b}{CDA'O} : \frac{c}{DAB'O},$$

From the ratios of a', b', c', g', h', k' , or, $a' : b' : c', a' : g' : k'$ respectively we deduce

$$CD BCO (b) CO A D A D O (b) D A O A B O (a) A B O,$$

$$C D A B O (b) A B O (a) A D A O A B O (b) A D A O,$$

$$B D A B O (b) A B O (a) B C D O (b) B C D O B C D O,$$

$$B C D (b) O (a) B C O (b) B C O (b) B C O (b) D A O D A O,$$

each of which equations represent a curve and admitting that it can be shown that these pass through O, G, G , &c., respectively, the equations are

$$B A B O (0) O (b) (a) P A A B A B A B,$$

$$B C D O (0) B C O (b) (a) B C D B C O B C O,$$

$$B D A O (0) D A O (b) (a) D A O D A O D A O,$$

$$D A B O (0) O (b) (a) D A B D A B D A B.$$

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7. Now the locus of P is evidently a curve, and this can only happen in the case where the four last-functions contain a common factor; here at first they have not this; for an arbitrary ABO if the requirement is $ABCO$, the conditions that the cubics suppose the these matters $ABCDOO$, the four extraneous factors being DAB , DAC , DAC , etc., and $ABCD(OO) = 0$, will be a cubic carry passing through the two points, and the like for $DAB = 0$. To verify in a particular instance, in B, C, D, A , are in a plane, then the conditions of agreement is $ABCDD(0)0 = 0$, that is, Δ_{ABC} , the conditions that the cubics

$$DABCO DABCO DABCO DABCO DABCO = 0$$

should pass through O, G, G . But then if the cubic passes through any a points of a cubic curve, it would in general be necessary that the cubic contained a common factor; here the cubic which contains a common factor, and a A axis, four common points; this is not at all impossible.

Correspondence, A, B, C, D and P and A', B', C', D' and P' subtending equal angles, and $AP, A'P'$ in a given ratio.

8. Consider, as before, A, B, C, D and P and A', B', C', D' and P' subtending equal angles, and the points P, P' being moreover such that the distances $AP, A'P'$ are in a given ratio. I write therefore

$$ar' : c''' : y' : z' = -\sigma a : b : c : d, \quad \text{where } a^2 = (1 - a^2 d)U = \Delta(0 - \cot C) V_3, \quad U' = \Delta(\cot C - \cot D) V_3, \quad U' =$$

respectively. We have

$$\begin{aligned} (AP)^2 &= a^2 y + (a^2 - c^2)(\beta x + (a^2 - c^2)y(x + a^2 - 2c^2y), \\ &= -a^2 P^2(a^2 z + (a^2 - c^2)x + (a^2 - c^2)y(x + a^2 - 2c^2y), \end{aligned}$$

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or, what is the same thing,

$$\frac{(AP)^2}{(1 - \alpha y - \beta x)^2 + Y(x + a^2 \cot A_2 y)};$$

and similarly

$$\frac{(A'P')^2}{(1 - \alpha y - \beta x)^2 + Y(x + a^2 \cot A_2 y)};$$

The required relation therefore is

$$\frac{a''(b'' + b'^2)V_a(a + a''^2 a'^2)}{(c + b'')^2(c + a''^2 + b'^2)} = \kappa^2 \frac{c^2(b'^2 + b'^2)V_a(a + a^2 + b'^2)}{(b - b')^2(c + a^2 + b'^2)};$$

viz. substituting for x' , y' , z' , a' their values

$$U = \Delta(0 - \cot C) V_3, \quad U = \Delta(\cot C - \cot D) V_3, \quad U = \Delta(\cot C - \cot C') V_3,$$

and similarly $U = \Delta(\cot C - \cot C') V_3, \quad U = \Delta(\cot C - \cot C') V_3,$

$$\beta'^2 = \beta'^2 \cot(C - \beta' \cot(b + a'^2 \cot(a'^2 + a''^2))(a''^2 + a''^2(b'^2 - a''^2)));$$

and the equation becomes

$$(BCD)Y(B'D)^2(B'A''CO) = (1 + B'D)^2(CO + AVCO) \\ + DAB'D(B'D)^2(B'D'')^2 \cdot A = (\beta^2 \beta'^2)(b'^2 \cot C - a''^2 \cot(C'(C - C')));$$

and the like equation for b .

The two Tetrahedra $(A, B, C, D$ or P and A', B', C', D' or P' in $ABC'D'$ subtending equal angles.

10. I consider now the lions-mentioned problem of the two tetrahedra; viz. on the two faces ABC and $A'BC'$ respectively, letting fall the perpendiculars DK and $D'K'$, then first A, B, C, D and A', B', C', D' subtend equal angles, the conditions for this is on a line, the focus of P is a cubic curve through A, B, C (ABC); and from the formula

$$F \cdot Y = K(DA' \cdot A'BO + D'A' \cdot A'BO)$$

derived from the points A, B, C , we have $BO = O' = CO, G' = G$, etc.

Next, B, C, D and P and B', C', D' and P' subtend equal angles, the conditions whereby distances AP and $A'P'$ are in a given ratio; the locus of P is a 12-line curve through $(BCDO)$.

[513]

On a Bicyclic Chuck.*[From the Philosophical Magazine, vol. XLIII. (1872), pp. 365–367.]*

p. 209. THE apparatus, although I have called it a chuck, is constructed not for turning, but for drawing: viz. it rotates horizontally, on a table (being moved, not from the inside by the axis of the lathe, but from the outside by a handle-frame), carrying a drawing-board which works under a fixed pencil supported by a bridge. Two points of the drawing-board describe circles, and the curve traced out on the drawing-board is consequently that described by a fixed point upon a moving plane (two points of which describe circles), or, what is really the same thing, the path of any point on a rigid body which has a sliding-piece (6) carrying a block (3), and besides a fixed plate by a handle-frame: which two are sides along (one of a -pieces of which are equal and respectively round the axis of the moving plane).

The apparatus consists in the first instance of a moving plane Q , sliding round on the surface of an horizontally rotating board. It is essential to remark that in this simple form of the instrument the chuck is a description of the chuck X , X are themselves the planes II, II, then putting the mitre wheel of one plane X in gear with either of the corresponding central wheels, and making the other rotate, the other plane X will rotate as well; according as the mitre wheels are in the opposite direction, or it will be in motion, according as the mitre wheel is in gear with one or the other of the corresponding central wheels, and may rotate with each of them.

To convert the apparatus into an oval chuck, we remove altogether the carrying-frame; and in the third plane we fix to the sides (8) of the handle-frame two bars at right angles to these sides, by means of pegs on the lower surfaces of these bars fitting tightly into holes on the sides (8) (which holes and the ends of the bars are shown in the figure), in such wise that these bars are have a moving piece (1) in the form of a rectangle with a circle cut out thereof, resting about the segments (4), and having upon it grooves in which works a sliding-piece (6) carrying a block (3); there is in this block a circular hole, D . The second plane includes also two sides (8) of a handle-frame, which two sides slide along two of the sides of the piece (1).

Third plane consists of a rectangular piece (9) rotating about an axis fixed to the block (3), and having a sliding-piece (10) in which is a circular hole, D . The third plane includes also the betoxed-mentioned block (7), having upon it the hole E .

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and it includes also the remaining two sides (11) of the handle-frame, and, let into the same so as to be flush therewith on the upper surface, two slips (12) completing, in this plane, the handle-frame.

We have thus on a level the sides (11), (12) of the handle-frame and the holes C , D , where C rotates about the point B , which is the centre of the block (3); and D rotates about the point A , which is the centre of the segments (4). Each hole being capable of describing a complete circle; and the distances AB , BC , CD , and DA are (within limits) adjustable to any given values: the distance of the holes C , D is made equal to that of the two pegs not referred to.

Connected herewith by means of cylindrical pegs working in the holes C , D respectively, we have a carrying-frame; viz. the fourth plane contains two sides (13) of this carrying-frame, and two moveable bars (14), attached to the remaining two sides (15) of the carrying-frame, and having on their lower surfaces the pegs which work in the holes C and D respectively—such bar being free to rotate about one extremity, and being clampable at the other extremity so as to allow the two pegs to be adjusted at a given distance from each other. And then in the fifth plane we have the remaining two sides (15) of the carrying-frame.

Rigidly connected with the carrying-frame we have the drawing-board; or, to make the whole more complete, this should be adjustable to any given position in regard to the carrying-frame by giving it two sliding motions crosswise, and a rotating motion, in the manner of an eccentric chuck.

To convert the apparatus into an oval chuck, we remove altogether the carrying-frame; and in the third plane we fix to the sides (8) of the handle-frame two bars at right angles to these sides, by means of pegs on the lower surfaces of these bars fitting tightly into holes on the sides (8) (which holes and the ends of the bars are shown in the figure), in such wise that these bars include between them the piece (9), which is thereby kept in a direction at right angles to the sides (8), and so slides between them. And we then take away the four points A and B respectively, by connecting the drawing-board and the third plane by a sliding-piece (16) carrying a handle-frame, it is easy to arrange as to this.

It is hardly necessary to remark that the pencil should have two sliding motions crosswise, so as to allow it to be adjusted to any given position; and a small up-and-down motion, so that it may be loaded to press with the proper force upon the drawing-board.

The variety of forms, even with a fixed adjustment of the chuck, only the position of the pencil being altered, is very considerable: among them we have lentil ovals and pear-shapes, passing through cusped forms into bent figures-of-eight.

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[Figure: Plan and elevation of the bicyclic chuck, three planes labelled — omitted]

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On the Problem of the In-and-Circumscribed Triangle.

[From the Philosophical Transactions of the Royal Society of London, vol. CXLII. (for the year 1871), pp. 369–412. Received December 30, 1870—Read February 9, 1871.]

p. 212. THE problem of the In-and-Circumscribed Triangle is a particular case of that of the In-and-Circumscribed Polygon; the last-mentioned problem may be thus stated:—to find a polygon such that the angles are situate on n given curves, and the sides on m given curves. And let us say in the first instance that as the number of such sides respectively are all of them distinct curves; then, considering the side AB as having upon it B , A , and another point, that is, as having upon it three points of the system; and analogously BC as having upon it three points, and so on, we see that the order of the curves must be such that the sides may be in succession to those of the points; that is, $n + m$ points, which is what each side must be in succession to those of the angles. The supposition that a on the polygon are not always such that they shall serve to find all of them, viz. a side being touched by the next side, the curves of any side may be excluded from the requirement of containing the the same angle of the polygon: the solution of the problem when the angles are on the same curve, and when the sides on a single one, viz. the case in question, in which the number of curves is $= 6$, and which has been previously considered; and there exists in (1871) by M. Poncelet a complete and elaborate discussion of the case where there is a curve common to the angles and sides, viz. in the case of the triangle is in (1817) referred to as $aDC[c[c[D]]]$, viz. a, b, c are respectively. That in A, D, F are the sides on, or as respectively, and a, b, c are the angle of FE, BD, DF .

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respectively. And I use the same letters a, c, e, B, D, F to denote the curves containing the angles and touched by the sides respectively; viz. the angle a is situate in the curve a , then B touches the curve B , and so on for the other angles and sides respectively. An equation such as $a = e$ or $a = B$ denotes that the curves a, c , or, as the case may be, the curves a, B are one and the same curve: it is in general convenient to use a new letter for denoting these identical curves; viz. I write, for instance, $a = e$ or $a = B$ as the case may be, the curves a, B are one and the same curve: it is in general convenient to use a new letter for denoting these identical curves; viz. I write, for instance, $a = e$ or $a = B$ as the case may be, the curves a, B are one and the same curve: it is in general convenient to use a new letter for denoting these identical curves; viz. I write, for instance, $a = e$ or $a = B$ as the case may be, the curves a, B are one and the same curve. The expression “no identities” denotes that the curves are all distinct. But I use also the same letters a, c, e, B, D, F to denote the curves containing the angles and touched by the sides respectively; viz. the angle a is situate in the curve a , B touches the curve B , and so on for the other angles and sides respectively; viz. to denote the orders and classes of the curves a, c, e, B, D, F, x, y quantitatively; thus the equation such an $aBcDD = 0$ denotes that the curves of aB, C, D are no identities; that is, the order of a are 0 and 0, those of B, D, F are 0 and 1, respectively.

The case “two several distinct identities” denotes that the curves a, c, e contain other, and selecting one or two of them, $a = c$ for “two identities” denotes the number of triangles is given as the before-mentioned result for the polygon. In the case $a = aBCDDF$, the methods and principles, however, are applicable to the case of polygon of any number of sides, and although consideration of the manner in which the consideration leads to the problem of the polygon, viz. to the present case we have here (AB , etc.), the relations of which are derived from this different square: roots and their results $32 = 6.4 \times C$ etc. But it is for further nature.

I remark that the work, in the following Table, of the number of positions of a particular angle of the triangle, and that in some cases, on account of the symmetry of the figure, the number of triangles is a submultiple of this number; viz. the number of positions of the angle is to be divided by 2 or 6; this is expressly shown, by means of a separate column, in the Table.

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Table: number of positions of an angle and number of triangles for various identifications among the six curves a, c, e, B, D, F .

NO.	IDENTIFIES	SPECIFICATION	SAMPLE	NO. OF TRIANGLES
1	$a = c = e$	$a = c = e$	$2\alpha\beta DF$	$2\alpha\beta DF$
2	0	no	$2\alpha\beta DF$	$2\alpha\beta DF \div 6$
3	2	$a = c \neq e$	$2EF \cdot D = 1 \text{ flex.}$	$2EF \cdot D - 1$
4	2	$a = c, B = D$	$1EF \cdot EF$	$\alpha\beta DF$
5	3	$a = c = e \neq B = D = F$	$1\beta(\alpha - 1)EF$	$1\beta(\alpha - 1)EF$
6	2	$a = c, B = D, e = F$	$2\alpha EF^2$	$2\alpha EF^2$
7	3	$a = c = e \neq B = D \neq F$	$1\beta(\alpha - 1)EF$	$1\beta(\alpha - 1)EF$

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[Table continues with rows 8–15: further identifications, specifications, samples and triangle counts. Tabular structure preserved; full numeric entries occupy several columns and are partially abbreviated.]

NO.	IDENTIFIES	SPECIFICATION	SAMPLE	NO. OF TRIANGLES
8	2	$a = c, e = D$	$2\alpha\beta EF$	$2\alpha\beta EF$
9	3	$a = c, B = D, e = F$	$E(\alpha - 1) \cdot DF$	$E(\alpha - 1)DF$
10	3	$a = c, e = B = D$	$2EF \cdot \alpha\beta DF$	$2EF \cdot \alpha\beta DF$
11	3	$a = e, B = D, e = F$	$\alpha\beta(\alpha - 1)DF$	$\alpha\beta(\alpha - 1)DF$
12	3	$a = c = e$	$1EF \cdot 1 \text{ flex.}$	$1EF \cdot 1$
13	4	$a = c, e = B = D, e = F$	$E\alpha\beta DF$	$E\alpha\beta DF$
14	4	$a = c = e, B = D, e = F$	$\alpha\beta(\alpha - 1)DF$	$\alpha\beta(\alpha - 1)DF$
15	4	$a = c = e$	$F \cdot B - D, e = F$	$F \cdot B - D$

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Table (continued): rows 16–23 with further identifications and specifications.

NO.	IDENTIFIES	SPECIFICATION	SAMPLE	NO. OF TRIANGLES
16	4	$a = c, B = D, e, F$	$2\alpha EF \cdot DF$	$2\alpha EF \cdot DF$
17	4	$a = c, B = D, e = F$	$2\alpha EF^2$	$2\alpha EF^2$
18	4	$a = c, e, B, D, e = F$	$\alpha\beta EF^2$	$\alpha\beta EF^2$
19	4	$a = c = e$	$\alpha = e$ flex.	$E\alpha$
20	4	$a = c, B = D$	$\alpha\beta EF$	$\alpha\beta EF$
21	5	$a = c, B = D, e = F$	$4\alpha\beta EF$	$4\alpha\beta EF$
22	5	$a = c, B = D, e = F$	$\alpha = e$	$E\alpha$
23	5	$a = c = e$	all equal	—

[End of paper 514 and of the section pp. 167–216 typeset here.]